

WENXIAO ZHANG

PhD Candidate in Computer Science · AI Engineer · Data Scientist

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EDUCATION

The University of Western Australia (2026 QS Rank: 77) 2024/03 – 2027/03

PhD in Computer Science

The University of Western Australia 2021/02 – 2023/07

Master of Software Engineering WAM 80.42

Xiamen University of Technology 2016/09 – 2020/06

Bachelor of Electronic and Information Engineering WAM 85.69

EXPERIENCE

Telstra Data Science Specialist 2024/11 – 2025/07

- Designed and shipped an enterprise **GraphRAG and agentic retrieval system** for frontline customer service: Neo4j graph on Docker over Salesforce data exposed via APIs; engineered agentic retrieval (query decomposition, multi-step planning, graph-grounded answers) to reduce enterprise hallucination.
- Built an **automated RAG evaluation system (AutoEval)** with retrieval and answer-faithfulness metrics so system value was measured, not assumed; delivered on Azure ML with Docker, CI/CD and TDD, accelerated by Claude Code and Codex.

The University of Western Australia Academic Tutor 2024/02 – Present

- Facilitated labs for **Cloud Computing** (AWS), **Intelligent Agents**, **Web Development** (Git/GitHub full-stack), **Cybersecurity** (CTF), and **Database Systems** (SQL) across large, mixed-ability cohorts.
- Mentored students through design, implementation and debugging; revised teaching materials, marked assignments and exams, and gave real-time feedback.

China Telecom Software Developer 2021/11 – 2022/02

- Rebuilt CMS frontend with **Vue.js**, delivering faster load times, modular components and maintainability.
- Designed RESTful APIs (**Spring Boot**, **MyBatis**), optimised SQL, collaborated with product and UI/UX.

Gillion Technologies Software Developer 2019/12 – 2020/05

- Maintained enterprise **maritime transport management system** for multiple clients with high availability.
- Built scalable frontend and backend solutions; set up **CI/CD** pipelines for automated delivery.

RESEARCH

Research Direction: Reliable and Secure Large Language Model Agent Systems for Complex Knowledge and Execution Tasks

1 CRAFT: Calibrated Reasoning with Answer-Faithful Traces for Multi-Hop Question Answering (First Author, EMNLP 2026, under review, **CORE A***)

- Addressing reasoning-answer inconsistency in multi-hop QA, proposes a framework producing **answer-faithful, auditable reasoning traces** via RL (GRPO) with deterministic + LLM-judge reward calibration.
- Key tech: Retrieval-Augmented Generation (RAG), Group Relative Policy Optimization (GRPO), faithfulness calibration, process supervision.

2 PRISMA: Reinforcement Learning Guided Two-Stage Policy Optimization in Multi-Agent Architecture for Open-Domain Multi-Hop Question Answering (First Author, TKDE 2026, under review, **CORE A***)

- For open-domain multi-hop QA, proposes the **PRISMA (Plan–Retrieve–Inspect–Solve–Memoize)** multi-agent retrieval architecture, achieving reasoning-guided retrieval via planning, inspection and memoization.
- Key tech: multi-agent reinforcement learning, agentic retrieval, multi-hop reasoning, memory mechanism.

3 *MIRAGE: How Conversation State Shapes Historical Evidence Use in Multimodal Personal Agents* (First Author, ACM MM 2026, under review, **CORE A***)

- Empirical study of **multimodal personal agents**: how **conversation state** shapes historical evidence use, revealing patterns of evidence loss and reasoning drift in long-running conversations.
- Key tech: multimodal personal agents, conversation state, historical evidence attribution, evaluation.

4 *CARE: Pre-Execution Command Verification for Shell-Executing LLM Agents* (First Author, ISSRE 2026, under review, **CORE A**)

- Targeting **LLM agent shell-execution safety**, proposes **CARE**: a **pre-execution verification architecture** using deterministic evidence (syntax, semantics, paths, risk) with an LLM judge for borderline cases, substantially reducing harmful executions at millisecond latency.
- Key tech: runtime assurance, LLM agent safety, shell command verification, static-LLM adjudication.

5 *STIndex: A Context-Aware Multi-Dimensional Spatiotemporal Information Extraction System* (First Author, WWW 2026, **CORE A***)

- Built an **end-to-end spatiotemporal information extraction system** using LLMs to extract coordinates from unstructured text, with an interactive dashboard providing a unified reference for downstream RAG and situational awareness.
- Key tech: spatiotemporal extraction, context-awareness, demand-driven indexing, visualisation.

6 *Docs2KG: Unified Knowledge Graph Construction from Heterogeneous Documents Assisted by Large Language Models* (Co-author, WWW 2025, **CORE A***)

- Proposes a **schema-agnostic** LLM framework that constructs structured knowledge graphs from large-scale heterogeneous documents, providing the knowledge foundation for GraphRAG.
- Key tech: knowledge graph construction, LLM information extraction, GraphRAG.

7 *Enhancing Reliability in LLM-Integrated Robotic Systems: A Unified Approach to Security and Safety* (First Author, JSS 2026, **CORE A**)

- Unified framework enhancing robustness of LLM-integrated systems under **prompt injection attacks**, strengthening the security and reliability of LLM-driven robotic systems.
- Key tech: LLM safety, prompt injection defense, system robustness.

8 *OpenOmni: A Collaborative Open Source Tool for Building Future-Ready Multimodal Conversational Agents* (Co-author, EMNLP 2024, **CORE A***, **Best Demo Paper**)

- Open-source **multimodal conversational agent** integrating STT/TTS, emotion analysis, RAG and LLMs; awarded EMNLP 2024 Best Demo Paper.
- Key tech: multimodal interaction, speech and emotion analysis, RAG, open-source engineering.

SKILLS

- **Languages:** Mandarin (native), English (fluent)
- **Agentic / LLM:** LangChain, LangGraph, MCP, OpenClaw, RAG, vLLM, harness engineering
- **Engineering:** Python, TypeScript, SQL/Cypher, Django, Docker, Neo4j, PostgreSQL, React/Next.js
- **Cloud & Delivery:** AWS, Azure ML, REST/microservices, CI/CD, Git, TDD, agile
- **AI-Assisted Development:** Claude Code, Codex (design, code generation, code review)